Akoizo Original Practice - 2027 Rotation Form 4

Division C | Codebusters | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (400 points) Decode the timed-base Aristocrat: NKVFI IVNJIEZ BFTV IVZHKQZ IVAIJEHNLWKV

Plain alphabet: ABCDEFGHIJKLMNOPQRSTUVWXYZ
Cipher alphabet: FWNEVMDULCTKBSJARIZQHYPGXO. Each row means plaintext letter -> ciphertext letter. This is a supplied one-to-one monoalphabetic substitution; no letter maps to itself. Timing is practiced separately; this worksheet awards the base score only.

Answer: _____
_____
_____

2. (400 points) Decode the Patristocrat: DNFYW WFDSW EXPYM FWFXL NZXWF UWSEL DJCNF

Spaces were removed and groups of five added for display.
Plain alphabet: ABCDEFGHIJKLMNOPQRSTUVWXYZ
Cipher alphabet (K2, keyed and cyclically shifted): YCDEFGHIJKMNPQSUVWXZLABORT

Answer: _____
_____
_____

3. (400 points) Decode this Spanish Xenocrypt: LQY UWLFCY QFDFXJZY EYZSX DNYWSX

Plain alphabet: ABCDEFGHIJKLMNOPQRSTUVWXYZ
Cipher alphabet (K2, keyed and cyclically shifted): YCDEFGHIJKMNPQSUVWXZLABORT. Plaintext is Spanish; no accent marks are encoded.

Answer: _____
_____
_____

4. (400 points) Decode this Baconian message: ababa aabaa aabaa abbbb abbab abbba baabb aabaa baaba

Use the supplied modern 26-letter mapping:
A=aaaaa; B=aaaab; C=aaaba; D=aaabb; E=aabaa; F=aabab; G=aabba; H=aabbb; I=abaaa; J=abaab; K=ababa; L=ababb; M=abbaa; N=abbab; O=abbba; P=abbbb; Q=baaaa; R=baaab; S=baaba; T=baabb; U=babaa; V=babab; W=babba; X=babbb; Y=bbaaa; Z=bbaab. Spaces separate encoded letters; restore word boundaries from meaning. Crib READ/TEST/CHECK/KEEP, as applicable: the plaintext begins with KEEP.

Answer: _____
_____
_____

5. (400 points) Decode this Nihilist message: 27 45 76 46 25 53 59

Use key DATA, repeated. I/J share a cell. Coordinates are row then column. Encryption adds plaintext coordinate plus key coordinate as ordinary integers.
1: A B C D E
2: F G H I K
3: L M N O P
4: Q R S T U
5: V W X Y Z

Answer: _____
_____
_____

6. (400 points) Decode this 5x5 Checkerboard message: CZ AV AX BZ CW EY AW CY EX EW BY DY BX BV BY EV AZ AY CY EZ AZ CX DV DZ AV DW DY DX

Rows are labeled ABCDE; columns VWXYZ. Read each pair row first. Polybius grid rows in order:
1: A B C D E
2: F G H I K
3: L M N O P
4: Q R S T U
5: V W X Y Z. All row and column labels occur in the ciphertext.

Answer: _____
_____
_____

7. (400 points) Solve this complete-columnar transposition: ATQEEAYX SDRIAUAI RAUCFNSX TAERRLLS

Four columns were filled left-to-right by rows; whole columns were read in an unknown order. Trailing X characters pad the last row. The plaintext begins with crib STAR at its first four positions. Recover the complete plaintext including padding X characters. There are no omitted cells.

Answer: _____
_____
_____

8. (2 points) A Homophonic Cipher keyword has four letters. Which property must those letters satisfy under the 2027 C rules?

Answer: _____
_____
_____

9. (2 points) A timed Aristocrat is solved correctly in 240 seconds. Compute the timing bonus using 2(600-t).

Answer: _____
_____
_____

10. (2 points) A 400-point plaintext answer has five character errors. Two errors are free and each further error costs 100 points. Find its score.

Answer: _____
_____
_____

11. (2 points) A 500-point keyword answer has three character errors. No keyword errors are free and each error costs 100 points. Find its score.

Answer: _____
_____
_____

12. (2 points) A proposed Aristocrat alphabet maps plaintext A to ciphertext A. Why is it invalid under the 2027 C rules?

Answer: _____
_____
_____

13. (2 points) A Fractionated Morse problem supplies a crib but does not identify its ciphertext placement. What information is missing under the 2027 C rules?

Answer: _____
_____
_____

14. (400 points) Decode this Fractionated Morse message: DUFJZEWPFDZHOPSI

A Morse letter separator is x and a word separator is xx; trailing x is padding. The plaintext begins with READ at its first four letters. Supplied fractionation table: A=...; B=..-; C=..x; D=.-.; E=.--; F=.-x; G=.x.; H=.x-; I=.xx; J=-..; K=-.-; L=-.x; M=--.; N=---; O=--x; P=-x.; Q=-x-; R=-xx; S=x..; T=x.-; U=x.x; V=x-.; W=x--; X=x-x; Y=xx.; Z=xx-. Morse letters: A=.-; B=-...; C=-.-.; D=-..; E=.; F=..-.; G=--.; H=....; I=..; J=.---; K=-.-; L=.-..; M=--; N=-.; O=---; P=.--.; Q=--.-; R=.-.; S=...; T=-; U=..-; V=...-; W=.--; X=-..-; Y=-.--; Z=--...

Answer: _____
_____
_____

15. (400 points) Decode this Porta message: MSAXYKPEEFTLE

Given keyword SCIENCE, repeated over letters only. A=0..Z=25. Each key pair selects a reciprocal substitution row. Upper alphabet for every row is ABCDEFGHIJKLMNOPQRSTUVWXYZ; lower rows: AB=NOPQRSTUVWXYZABCDEFGHIJKLM; CD=OPQRSTUVWXYZNMABCDEFGHIJKL; EF=PQRSTUVWXYZNOLMABCDEFGHIJK; GH=QRSTUVWXYZNOPKLMABCDEFGHIJ; IJ=RSTUVWXYZNOPQJKLMABCDEFGHI; KL=STUVWXYZNOPQRIJKLMABCDEFGH; MN=TUVWXYZNOPQRSHIJKLMABCDEFG; OP=UVWXYZNOPQRSTGHIJKLMABCDEF; QR=VWXYZNOPQRSTUFGHIJKLMABCDE; ST=WXYZNOPQRSTUVEFGHIJKLMABCD; UV=XYZNOPQRSTUVWDEFGHIJKLMABC; WX=YZNOPQRSTUVWXCDEFGHIJKLMAB; YZ=ZNOPQRSTUVWXYBCDEFGHIJKLMA.

Answer: _____
_____
_____

16. (2 points) How do keyword-answer scoring and plaintext-answer scoring differ in their free-error allowances?

Answer: _____
_____
_____

17. (2 points) Why must a monoalphabetic substitution use a globally consistent mapping?

Answer: _____
_____
_____

18. (2 points) A timed puzzle is first solved after 650 seconds. What timing bonus remains?

Answer: _____
_____
_____

19. (2 points) Two special-bonus questions are solved with no penalty points. What aggregate special bonus does the 2027 C schedule award?

Answer: _____
_____
_____

20. (2 points) Which permitted monoalphabetic format removes word spaces and displays ciphertext in groups of five?

Answer: _____
_____
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